

99.500 Applied Mathematics for Engineering

Homework 2

February 25, 2026

1. A polymer film of thickness d is coated onto a metal substrate. The film's surface at $x = 0$ is exposed to air and irradiated by an ultraviolet (UV) source. This irradiation induces a photochemical cleavage reaction, producing monomers at a local volumetric rate of r_m with unit $\frac{\text{mol}}{\text{cm}^3\text{s}}$. The reaction can be represented as: $(M)_n \rightarrow M + (M)_{n-1}$, where $(M)_n$ is the polymer, M is the monomer, and $(M)_{n-1}$ is the polymer with one fewer monomer unit. The monomer is soluble in the polymer and, since the substrate is impermeable, it can only diffuse out of the film through the $x = 0$ surface. The depolymerization kinetics follow zero-order behavior with respect to the dissolved monomer concentration, but the reaction rate decreases with depth due to the absorption of UV radiation, following an exponential attenuation law as

$$r_m = r_{m_0} e^{-\lambda x} \quad (1)$$

where r_{m_0} is the production rate at the surface $x = 0$ and λ is the absorption coefficient that characterizes how strongly the polymer absorbs the UV light. Assuming Fickian diffusion governs the transport of the dissolved monomer and that a pseudo-steady-state condition applies, the governing equation and boundary conditions for the monomer concentration are given as

$$D \frac{d^2 C}{dx^2} = -r_m \quad (2)$$

$$C(0) = 0 \text{ \& \> } \left. \frac{dC}{dx} \right|_{x=d} = 0 \quad (3)$$

Solve the above ODE to obtain an explicit expression for the monomer concentration.

2. A thin, disk-shaped platinum (Pt) particle with radius R in cm is positioned on a Zirconium dioxide (ZrO_2) surface of infinite extent, and the entire system is exposed to a dioxygen (O_2) atmosphere. The Pt selectively chemisorbs dioxygen at its perimeter, forming oxygen adatoms. The ZrO_2 surface itself does not adsorb oxygen. The oxygen adatoms generated at the particle perimeter diffuse along the Pt/ZrO_2 interface and react with oxygen vacancies in the ZrO_2 crystal lattice. This reaction follows first-order kinetics with respect to the oxygen adatom concentration. The diffusion of oxygen adatoms along the Pt/ZrO_2 interface is governed by Fickian diffusion, with a surface flux N_s with unit $\frac{\text{mol}}{\text{cm s}}$ as

$$N_s = -D_s \frac{dC_s}{dr} \quad (4)$$

where C_s is the surface concentration of adsorbed oxygen in $\frac{\text{mol}}{\text{cm}^2}$ and D_s is the surface diffusivity constant in $\frac{\text{cm}^2}{\text{s}}$. Assume that the number density of oxygen-vacancy sites at the Pt/ZrO_2 interface remains constant, meaning that the ZrO_2 acts as an infinite sink for oxygen. Additionally, suppose that the adsorbed oxygen at the disk perimeter is in equilibrium with the gas phase, leading to a known adsorbed oxygen concentration at radius R , denoted as C_{s_0} . The governing equation for the oxygen adatoms and its boundary conditions can be derived as

$$r^2 \frac{d^2 C_s}{dr^2} + r \frac{dC_s}{dr} - r^2 \frac{k}{D_s} C_s = 0 \quad (5)$$

$$C_s(R) = C_{s_0} \text{ \& \> } \left. \frac{dC_s}{dr} \right|_{r=0} = 0 \quad (6)$$

where k is the pseudo first-order rate constant for the reaction between oxygen adatom and oxygen vacancies.

(a) Solve the above ODE to obtain an explicit expression for $C_s(r)$. (ODE derivation is not required).

(b) The steady-state reaction rate of absorbed oxygen R_0 for the entire disk can be calculated by evaluating the flux at the surface, $R_0 = -2\pi RN|_{r=R}$. Calculate R_0 in terms of R .

(c) Sketch a plot of R_0 vs R and investigate the “small R ” and “large R ” behaviours. (Please include the slope and the asymptotic value of the plot)